SEMINAR SUMMARY:
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CUTTING KNOTS DOWN TO SIZE

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The goal behind the talk was to inspect tangle decomposition as a tool for understanding 3-manifold theory and $\mathbb{R}^3$ topology. We begin by considering a few definitions.

Definition 1.

i A *knot* is an embedding of a circle into $\mathbb{R}^3$ space or an embedding into a 3-sphere.
ii A *link* L is union of knots.
iii A *link diagram* is a projection of L into the plane with over-crossing and under-crossings explicitly labeled.
iv *Spacial graphs* or a *spacial representation* $\mathcal{R}(G)$, of a graph G, is the embedded image of G in $\mathbb{R}^3$ where the vertices of G are distinct points in $\mathbb{R}^3$ and the edges between them are simple Jordan curves between them in such a way that any two curves either are disjoint or they meet in a common end point.
v A *theta curve* (or θ-curve) is a spacial embedding of the theta-graph in $\mathbb{R}^3$ (specifically in the 3-sphere S^3 or Euclidean space $\mathbb{R}^3$).
vi *Ambient Isotopy* defines when two knots are considered equivalent by determining if one can be continuously deformed into the other within $\mathbb{R}^3$ without cutting, breaking, or passing the string through itself.
vii A 3-manifold is a topological space where every ball has a neighborhood homomorphic to a 3-dimensional Euclidean ball.

We then were given a list of problems of interest:

(1) Define and calculate invariants - to clarify, quantify, or tabulate invariants or to solve bigger problems.
(2) Studying unknotting operations - what subject is simplest? How is a structure formed?
(3) Explore connections - (a) apply 3-manifold topology to study knots; (b) apply knot theory to study 3-manifolds
(4) Apply topology problems in molecular biology (DNA and protien folding)

A list of tangle decompositions follow as:

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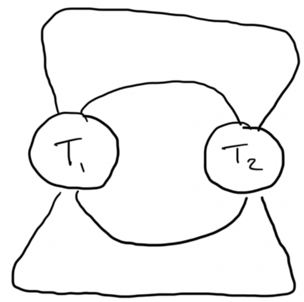

FIGURE 1. Basis for tangle tecomposition

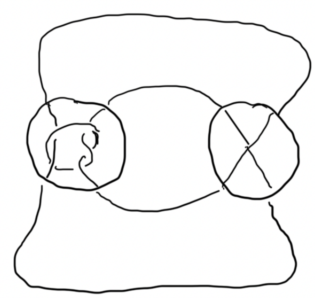

FIGURE 2. Example for tangle tecomposition

(1) Decoposition: $(S^3, K) = T_1 \cup T_2$
(2) Two-string tangle: $T_i = (\mathcal{R}]^{\ni}, t_i)$
(3) Conway sphere: $S \cap K \rightarrow$ 4-point sphere.
(4) Numerator closure: $N(T_1 + T_2) = S$

And, Figure 2 gives a classical knot example represented using tangle decomposition.

The Jones Polynomial. Using tangle decomposition and Figure 1, we get the following representations

$$\begin{aligned} N(T_1 + T_2) &= L \\ \langle T \rangle &= f_{\mathrm{T}}[0] + g_{\Gamma}[\infty] \\ \langle D_L \rangle &= \begin{bmatrix} f_{r_1} & g_{r_1} \end{bmatrix} \begin{bmatrix} \delta & 1 \\ 1 & \delta \end{bmatrix} \begin{bmatrix} f_{r_2} \\ g_{r_2} \end{bmatrix} \end{aligned}$$

A fancier technique. Use a tangle decomposition to "modularize" a complicated homology calculation. Algebraic invariants associated to tangles have geometric rationalizations aw immersed multi-curves. We use a strategy to calculate KH using a technique developed by Bar-Natan, and Kotelskiy-Watson-Zibrowius. Further, we use a strategy to calculate HFK by Zibrowius. Also, we consider the following tangle decompositions:

(1) Knot Floer homology: $\widehat{HFK}(K) = \bigoplus_i^n \mathbb{F}^d_{(m_i,a_i)}$
(2) Zibrowius's Peculiar Modules: $T \Rightarrow$ Heegaard diagram $\Rightarrow CFT^\delta \Rightarrow HFT(T)$
- Decorations involve bigradings, local systems
- Punctured sphere is parameterized
- This peculiar is a curved type D structure over an algebra - a type of Heegaard Floer module.

Theorem. (Zibrowius, The Pairing Theorem) Let $\Gamma_i = HFT(T_i)$ be the immersed curve representing the peculiar module of the tangle T_i. Then the Knot Floer homology of $K = T_1 \cup T_2$ can be obtained by the Lagrangian intersection homology on the 4-punctured sphere:

$$\widehat{HFK}(K) \otimes V \cong HF(\Gamma_1, \Gamma_2^*).$$

Theorem. (Lidman-Moore-Zibrowius, 2022) L-space knots have no essential Conway spheres.

We then considered branched covering space examples

(1) The wedge of two circles is a 2-fold cover of the circle branched over one point.
(2) The cylinder is a 2-fold cover of the disc branched over two points.
(3) $\Sigma_2(B^3, T)$: The double cover of a ball branched over a 2-string tangle is a 3-manifold with a torus boundry.
(4) $\Sigma_2(S^3, K)$: The double cover of a 3-sphere branched over a knot is a closed 3-manifold.

Theorem. (Monsanto's "Trick", 1976) Correspondence between rational Dehn fillings of the 3-manifold $\Sigma_2(B, T)$ and rational tangle fillings $N(B + T)$. (Some heavy drawings, have to refer to the slides here for the theorem).

We then spent time talking about the applications of Knot Theory regarding two biological applications, DNA and protien folding. We noted that Specific Site Recombinase is a knot with 15 crossings and is a well known standard knot in the literature.

We then proceeded to go over "The Tangle Calculus" which is a method for solving tangle equations due to Ernst and Summers (1980s) that relies on Dehn surgeries. The following set of equations was given

$$\begin{aligned} N(S+T) &= \text{substrate knot} \\ N(S+R) &= \text{product knot} \end{aligned}$$

Lastly we went into the idea of "unknotting-number" which has some quickly describable open problems associated with it and the decipline was pioneered by Scharlemann (for more reading).